

Game Carousel Onboarding & Automation – High Level Design

1. Purpose & Scope

This document outlines a unified process for automating the integration of new games into the BucStop game carousel. It provides both teams with a shared framework for game onboarding, ensuring consistency, speed, and error reduction by leveraging automation through manifest-driven workflows.

2. High-Level Integration & Automation

Goal: Add a new game to the carousel with one manifest + assets and let the pipeline do the rest.

Inputs - Game manifest (YAML/JSON): key, display name, ports, controller/namespace, feature flag, gateway/web entries. - Assets: 1 image (16:9), 1 JS bundle exporting `init()`.

Trigger - Open a PR adding the manifest (and assets) to the game registry repo.

Pipeline (CI) - **1)** Validate schema, uniqueness (gameKey, ID), and port conflicts. **2)** Verify assets (size, aspect, JS bundle signature). **3)** Auto-patch repos: - **Gateway:** add routes/URLs & health path. - **WebApp:** add carousel entry + feature flag entry. - **Service (optional):** scaffold/rename namespace, update GameInfo, appsettings, Docker.

Deploy (CD) - Build & deploy to **Dev** → **Stage** → **Prod** with approvals. - Run smoke tests; if healthy, **enable the feature flag** to show in carousel.

Runtime Data - WebApp emits **playcount & logs** to data store/dashboards for monitoring.

Result - Tile appears in carousel; clicking routes through **Game API GW** to the new game service.

3. High-Level Architecture Overview

3.1 Component Diagram

Dev adds Game Manifest → CI validates → Scaffolder patches repos → Deploy & Test → Enable Feature Flag

3.2 Component Breakdown

- **Registry & Validator:** Validates manifest syntax, IDs, ports, and assets.
 - **Scaffolder & Patcher:** Auto-modifies source code and configuration files.
 - **Gateway Updater:** Registers game with service URLs and routing.
 - **WebApp Updater:** Adds the game to the carousel and feature flag list.
 - **Asset Manager:** Validates and copies static files to the correct directories.
 - **CI/CD Pipeline:** Deploys updated components through environments (Dev → Stage → Prod).
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4. Validation Rules

- Unique gameKey & auto-incremented ID
 - Ports verified as available
 - Images must be ≤300KB, 16:9 aspect ratio
 - JavaScript bundles must export an `init()` function
 - Gateway entries require an active health endpoint
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5. Deployment & Rollback

- **Development:** Feature flag off by default; toggled on after successful smoke test.
- **Staging:** Manual QA approval; progressive rollout.
- **Production:** Manual approval; flag toggled on after validation.
- **Rollback:** Disable feature flag instantly; automated reversion of PRs by the pipeline.